Manual

Evaluations in the QM Sub System

QMS-AQS 8.1

Version 1.0.1374

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Copyright

Contents

1 Evaluations in the QM Sub System

Purpose

This component is used when the attributive and variable characteristics of inspection data have to be displayed in combined control chart and histogram format.

In order to use this function it is necessary to transfer the control chart type with a customer specific interface. The control chart type isn´t part of the standard interface QM-IDI from SAP-QM.

Implementation Considerations

This component is especially useful for short orders with few inspections since reliable quality information is only possible with sufficiently large data volumes. Its use is also recommended when the values have to be monitored over a certain period of time, e.g. to test the effectiveness of process improvements.

This component requires the inspection data to have been captured in HYDRA.

Integration

The component "SAP QM Quality Assurance Subsystem" with its associated inspection requirements and inspection results forms the base data, where the inspection data must have been previously entered through the AIP data entry client.

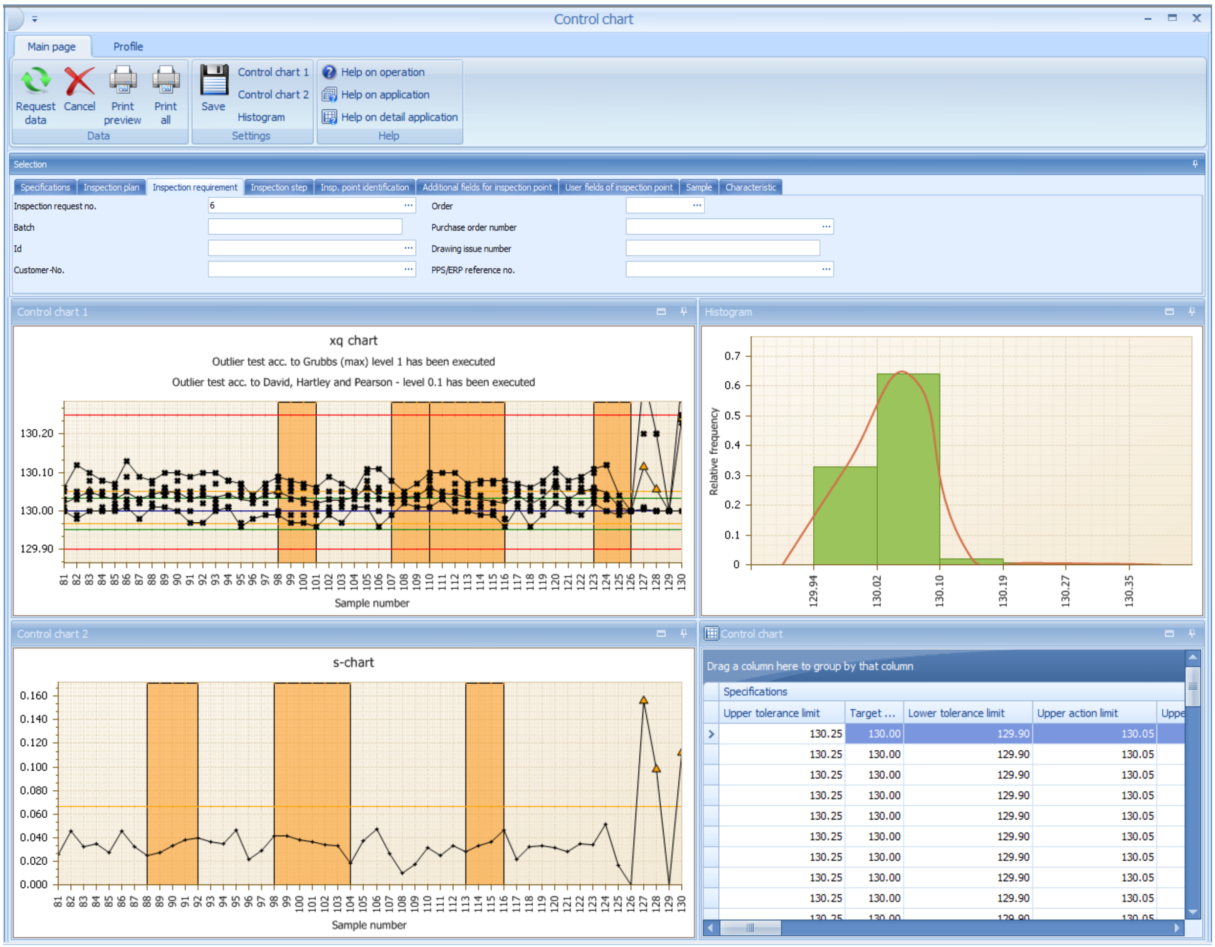
Features

Tested characteristics can be displayed graphically across orders in various control chart formats, as a histogram and in a list of the base data. The base data can be restricted through a wealth of available filters.

2 Control Chart Evaluation

Summary

Menu	Quality management → QM evaluation → Control chart
Transaction code	ccep
Function authorization	ccep



Utilization

Different control charts, the histogram and a data set list, which apply to several orders, are displayed for checked characteristics in the "control chart" report. The data basis may be restricted by using versatile filters.

Integration

Control charts may be evaluated on the basis of inspection data collected in the areas

- goods receipt,
- initial sample inspection,
- production and
- goods issue

Consequently, the "control chart" report is a comprehensive analysis tool.

Prerequisite

The below filter fields have to be filled out to be able to display control charts.

- Type ("specifications" tab)
- Area ("specifications" tab)
- OP sequence or characteristic number ("characteristic" tab)

Inspection order characteristics can only be identified uniquely if the operation sequence number is indicated. As the same characteristic number may be used several times within one inspection step, the inspection step characteristic cannot be identified uniquely even if the order number or yet the inspection request and inspection step number are added as further filter criteria. In this case, the computation of statistical key figures, e.g. cpk value calculation that is printed in a form, is normally suppressed, as the available tolerance limits are not necessarily identical for the whole, filtered data set. The inspection step characteristic can only be identified uniquely if the operation sequence number, inspection request number and inspection step number are filtered.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

"Specifications" tab

Type

Selection list of area types
- in-production inspection
- goods receipt inspection
- goods issue inspection
- initial sample inspection

The list of area types may be restricted subject to the respective licenses in use.

Area

Selection list of the configured areas of the previously filtered area type. By default, the following areas are available.

- In-production inspection: production
- Goods receipt inspection: goods receipt
- Goods issue inspection: goods issue
- Initial sample inspection: initial sample

Further areas can be generated through customizing.

"Inspection step" tab

Operation

Operation number

"Inspection point identification" tab

You require a new program version of this application if these fields are not available.

Workplace

Workplace number

Partial batch

Partial batch number

Batch

Batch number

Field 1…3

Additional fields for inspection points

"Additional fields for inspection points" tab

Field 4…8

Additional fields for inspection points

You require a new program version of this application if these fields are not available.

All fields included in this application do not support matchcode filtering for technical reasons.

Toolbar

Control chart 1 settings

Opens a dialog to configure the settings of control chart 1. The corresponding details are described in the respective detail application.

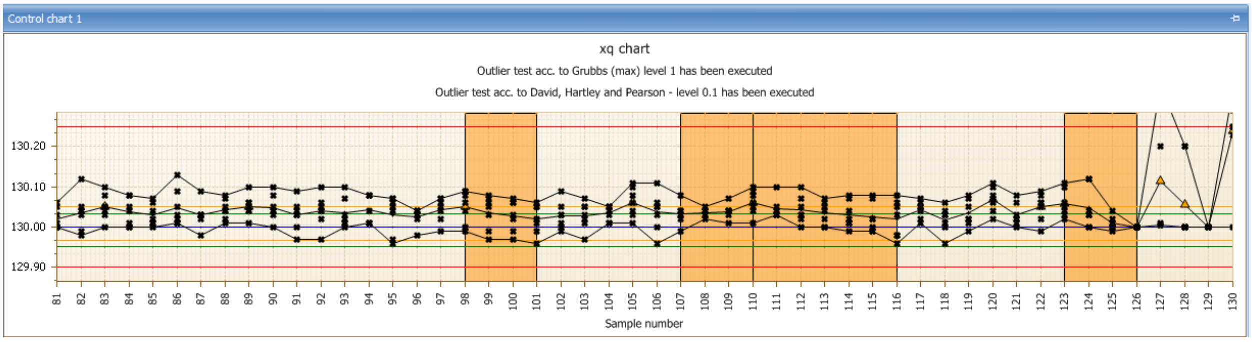
Control chart 2 settings

Opens a dialog to configure the settings of control chart 2. The corresponding details are described in the respective detail application.

Histogram settings

Opens a dialog to configure the histogram settings. The corresponding details are described in the respective detail application.

"Control chart 1" detail application



The contents of this application are defined by opening the dialog to configure "control chart 1". Changes made via this dialog are saved in relation to the user. Unless otherwise configured, the xq chart ("variable" characteristic type) or the p chart ("inspection chart" or "attributive" characteristic type) is displayed by default. Subject to the characteristic type, the user may choose to display one of the following control charts by default.

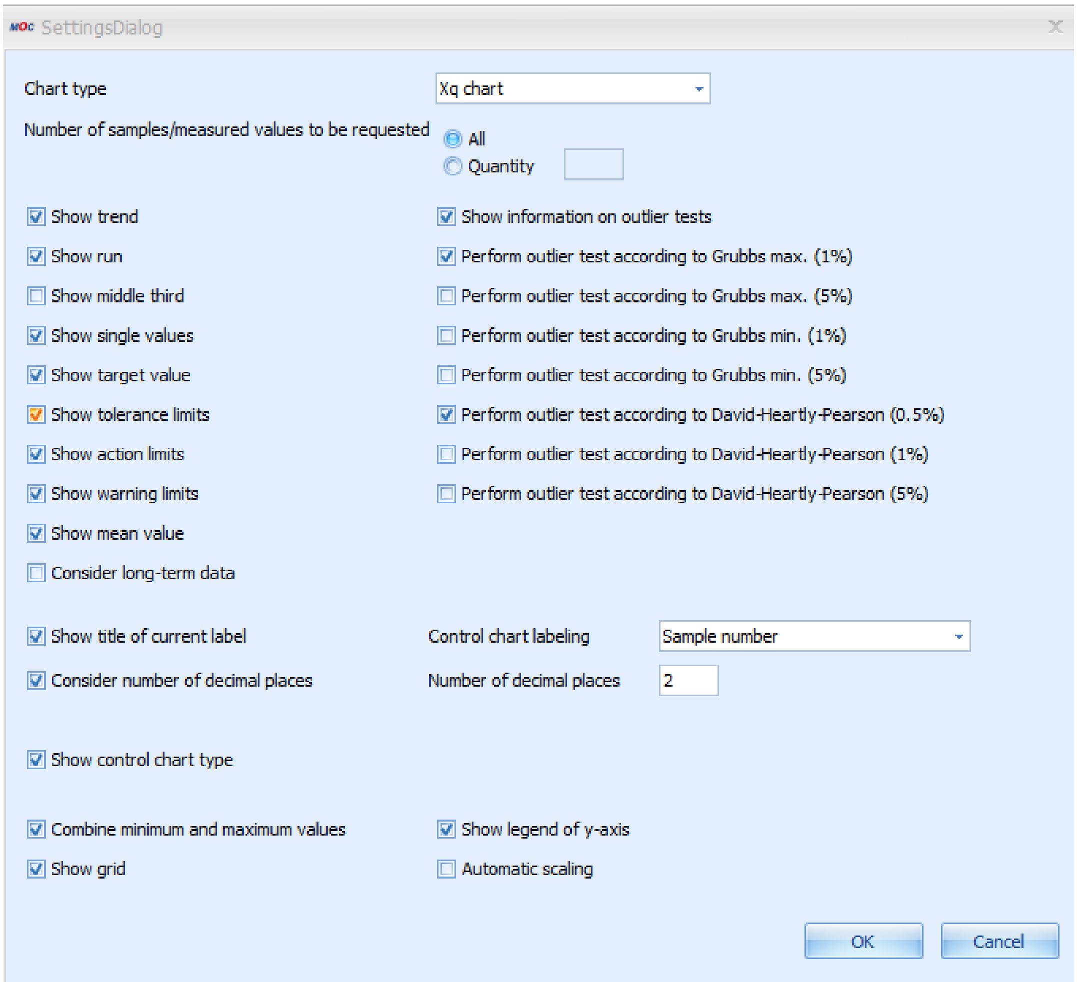
Variable characteristic

- Xq chart

- s chart
- R chart
- Single value chart
- Median chart

Attributive characteristic

- p chart
- np chart
- c chart
- u chart



The paragraphs that follow explain the essential configuration options.

Number of the samples to be requested

Specifies how many samples or single measured values are to be displayed in the control chart.

Display …

The options that include the term "display ..." enable or disable the presentation of the corresponding data in the control chart.

Consider long-term data

This option has to be checked to integrate archived data from the medium-term data area.

Combine minimum and maximum values

It is recommendable to show the minimum and maximum values that are each connected by a line to improve the presentation of the range of dispersion of single values within a sample.

Automatic scaling

The "automatic scaling" function allows for all values to be displayed, irrespective of the existing limit values. This shows even extreme outliers in the control chart. The disadvantage is that there is less space for the other measured values and, it is very often the case that a changing value pattern can hardly be recognized.

Show trend / run / middle third

The monitoring functions "trend", "run" and "MiddleThird" allow better surveillance of a process than by using the control chart alone. However, they can only be used with control charts for xq and median. The trend allows visualization of an upward or downward tendency in the process over several samples. By default, these are seven subsequent rising or falling values. The run shows sections in which the process runs above or below the mean value (when displayed, otherwise the target value) over several samples. By default a run is recognized if seven subsequent values are above the mean value. The number of seven samples/values for recognizing a trend or run, which is set by default, may be changed by customizing the system. "MiddleThird" refers to an unusually high or low number of values, in the section of the control chart viewed, within the middle third of the area bounded by the action limits.

Respective analyses are performed automatically for "trend", "run" and "MiddleThird". The control chart shows in a graphic if a trend, run and / or MiddleThird is available/detected. The following events are altogether highlighted by icons or color codes in the control chart.

- Trend (can be recognized by a colored area)
- Run (can be recognized by a colored area))
- Middle Third (can be recognized by a colored area)
- Outlier (icon: )
- Xq violates action limit (icon )

The presentation of outliers is restricted to the xq chart and median chart and connected with the presentation of single values. There are different functions for performing and presenting outlier tests. The different outlier tests for the different levels may be activated separately. Provided that the presentation of outlier tests has been activated, the result of this test is displayed in text form above the corresponding control chart.

The below outlier tests are available for the different inspection levels, whereas the inspection level is indicated in parentheses.

- Grubbs max. (1 %)
- Grubbs max. (5 %)
- Grubbs min. (1 %)
- Grubbs min. (5 %)
- David-Hartley-Pearson (0.5 %)
- David-Hartley-Pearson (1 %)
- David-Hartley-Pearson (5 %)

Notes on outlier tests

- The outlier tests do not refer to the collectivity of all samples, i.e. every sample is considered individually. Consequently, the following phenomena may appear:
  - Despite a large range between minimum and maximum value no outlier can be identified. A reason for this may be the equal distribution of the single values within the sample.
  - Despite a low range between minimum and maximum value an outlier can be identified. A reason for this may be an accumulation of many values at one “point” so that an individual value having a certain distance to this agglomeration is identified as outlier within the sample.
- At least three values have to be in the sample in order to be able to perform an outlier test. The more values are available within a sample the more uniform the general view of the outliers becomes compared to all samples.
- The Grubbs outlier test is performed with a sample size of $2 < n < 148$ (n = number of values within a sample). The outlier test according to David, Hartley und Pearson is performed at a sample size of $2 < n < 1251$ (n = number of values within a sample).

X-axis labeling

The below information is available to label the x-axis of control charts.

- Sample number

- Order number
- PPS reference number
- Purchase order number
- Batch
- Article number
- Machine number
- Cavity number
- Date + time of the first measured value of a sample
- Date + time of the last measured value of the sample
- Date + time of sample completion
- Badge number of the first measured value of the sample
- Badge number of the last measured value of the sample
- Badge number of sample completion
- Partial batch
- Workplace
- Production workplace
- Field 1
- Field 2
- Field 3
- Field 4
- Field 5
- Field 6
- Field 7
- Field 8

The fields "field 1" to "field 8" are enabled subject to MPDV customizing and are assigned an individual designation. As these field names are flexible, they are only entitled "field 1" to "field 8" in this document.
Field 1 includes, for example, the tool if cavity-related data collection is enabled..

Tool tips within the control chart

When a value is labeled being an outlier (red rhomb) detailed information on which test(s) was (were) the crucial factor for this determination is displayed when going with the mouse over this labeling (rhomb).

For mean values the tool tip shows the value, date and time for the first and last measured value of this sample as well as the corresponding inspection request number and inspection step number.

The exact value is shown for single measured values.

A special note is also displayed for the colored areas when a trend, run or middlethird is recognized. This note states the reason why this section has been colored.

Sorting of measured values

The type of control chart sorting is determined by the server and depends on how the control chart filters are configured. If the filters operation sequence and inspection request number or operation sequence and inspection step number are indicated the characteristic will be identified uniquely and sorting is based on the sample number.

Sorting is based on date and time if either the filter field "operation sequence" is left empty or it is filled out and the fields "inspection request number" and "inspection step number" are left empty instead.

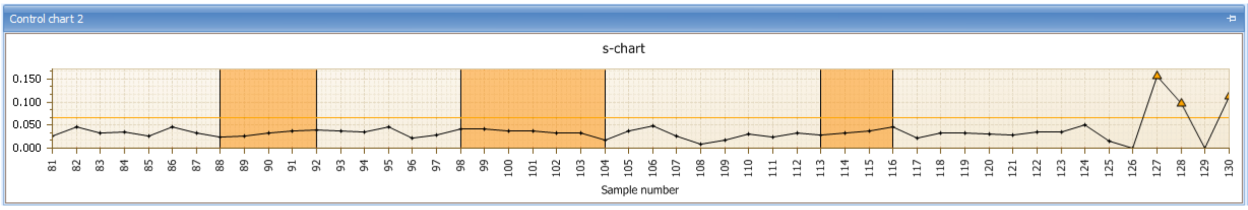
The optical presentation of trend or run always refers to how the measured values are sorted. However, the automatic generation of the failure type "trend" is always based on the measured values being sorted by the sample number, as the numbers of the operation sequence, inspection request and the inspection step are always known at the time when data is recorded.

Consequently, the following scenario might be possible.

- The control chart is sorted by date and time and a trend is shown, although no automatic failure "trend" has been generated.

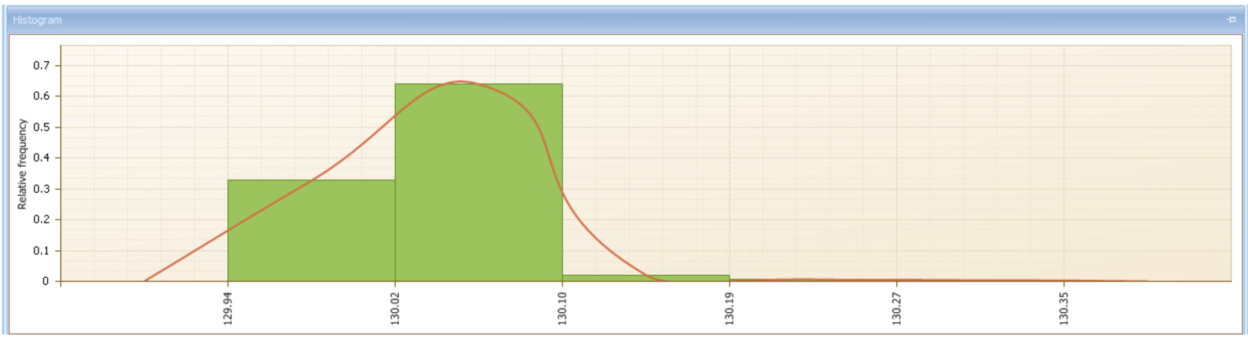
- The control chart is sorted by the sample number and no trend is shown, although the automatic failure "trend" has been generated.

"Control chart 2" detail applications

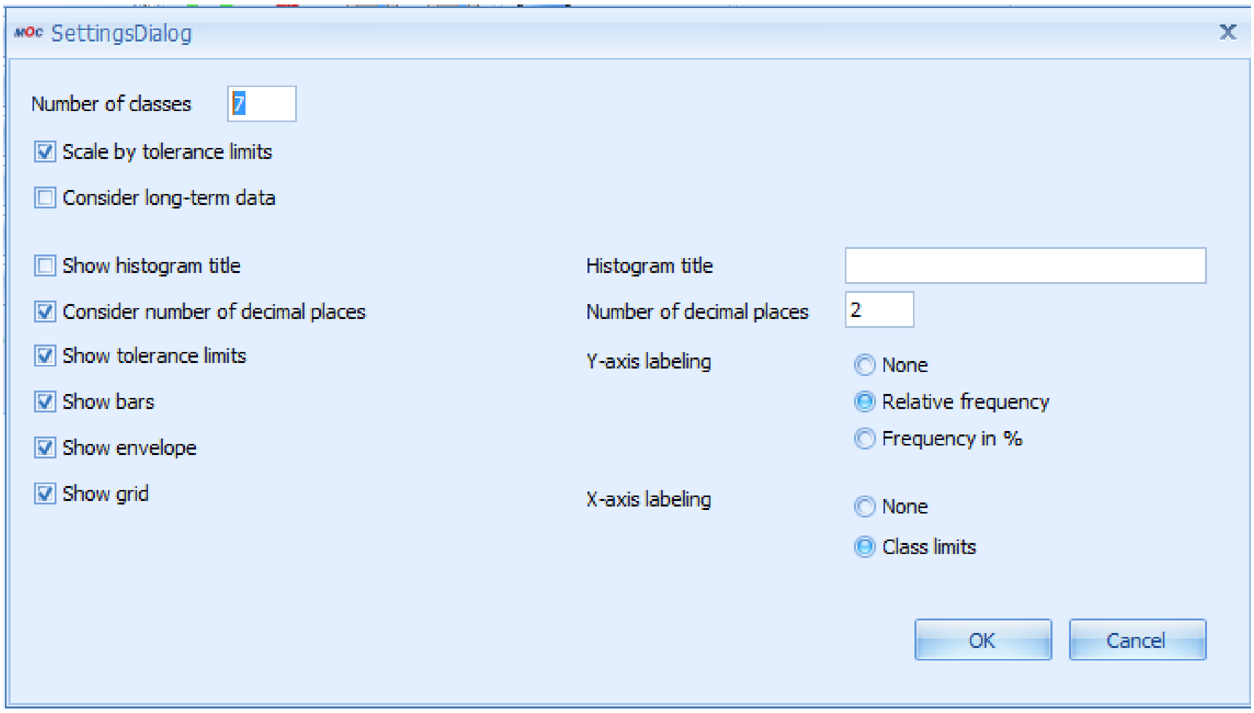


The features and configuration options of the "control chart 2" detail application are not explained here, as they match those of the "control chart 1" detail application.

"Histogram" detail applications



Unlike the control charts, which display a specified excerpt from the sample set, the histogram is always based on the entire set of available samples matching the selection filter criteria. The appearance of the histogram is determined by the number of classes and by any elements additionally displayed. The contents of this application are defined by opening the dialog to configure the "histogram". Changes made via this dialog are saved in relation to the user.



The paragraphs that follow explain the essential configuration options.

Number of classes

Determines the number of histogram classes according to which the measured values are to be distributed. If represented within the tolerance limits (option "scale by tolerance limits") one histogram class each is outside of the tolerance limits.

Scale by tolerance limits

Enabled: The classes are in between the tolerance limits with each one "outlier class" to the left and to the right.

Disabled: The classes include the range of all measured values (no separate classes for values outside of the tolerance limits).

Consider long-term data

Includes the archived data from the medium-term data area.

Show histogram title

If a special title is to be displayed it may be entered by enabling this option.

X-axis labeling

The x-axis shows the corresponding values of the class limits if the "class limits" option is set. The number of decimal places that are to be displayed may be set by the two configuration options for decimal places.

Consider the number of decimal places

Takes into account the defined number of decimal places in the x-axis labeling.

"Control chart 1 list" and "control chart 2 list" detail applications.


Control chart

Drag a column here to group by that column

Specifications								Main page		
Upper tolerance limit	Target value	Lower tolerance limit	Upper action limit	Upper warning limit	Lower action limit	Mean value	Lower warning li...	Date of first me...	Time of first me...	Date
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/13/
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/11/
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/10/
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/9/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/9/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2

The detail applications "control chart 1 list" and "control chart 2 list" show the data of control chart 1 and 2 in list form. If required, further analyses based on this data can be made using the Excel export function. The Excel export requires a special license.

This list does not show the user fields of inspection points for technical reasons.